PHASE 4

The Strategy Brain

How SamQuant turns old prices into disciplined portfolio decisions

1 **MARKET DATA** Validated OHLCV → 2 **STRATEGY** Study past closes → 3 **TARGET WEIGHTS** What % to own → 4 **BACKTESTER** Trade next open

THE RULE THAT PROTECTS THE WHOLE PROJECT

Monday close creates the signal -> Tuesday open executes the trade

ARCHITECTURE

One contract, three personalities

Every strategy speaks the same language: a target-weight DataFrame.

{AAPL: OHLCV, MSFT: OHLCV} → **validated_close_prices()** → **generate_target_weights()** → **Date x Symbol weights** → **Backtester.run()**

1 Validate once

Normalize symbols, reject broken OHLCV, and require every asset to use the same dates.

2 Decide only

A strategy says the desired percentage. It does not touch cash, fees, orders, or fills.

3 Execute later

The engine shifts decisions by one row and trades at the following opening price.

Why this wins in interviews: research logic can change without rewriting accounting logic. This is separation of concerns.

STRATEGY 1

Moving average: follow the trend

Question: is the recent direction stronger than the long-term direction?

BUY / HOLD
short average > long average

CASH
short average <= long average

WAIT
No long average yet

→

COMPARE
Short vs long

→

ALLOCATE
Split equally

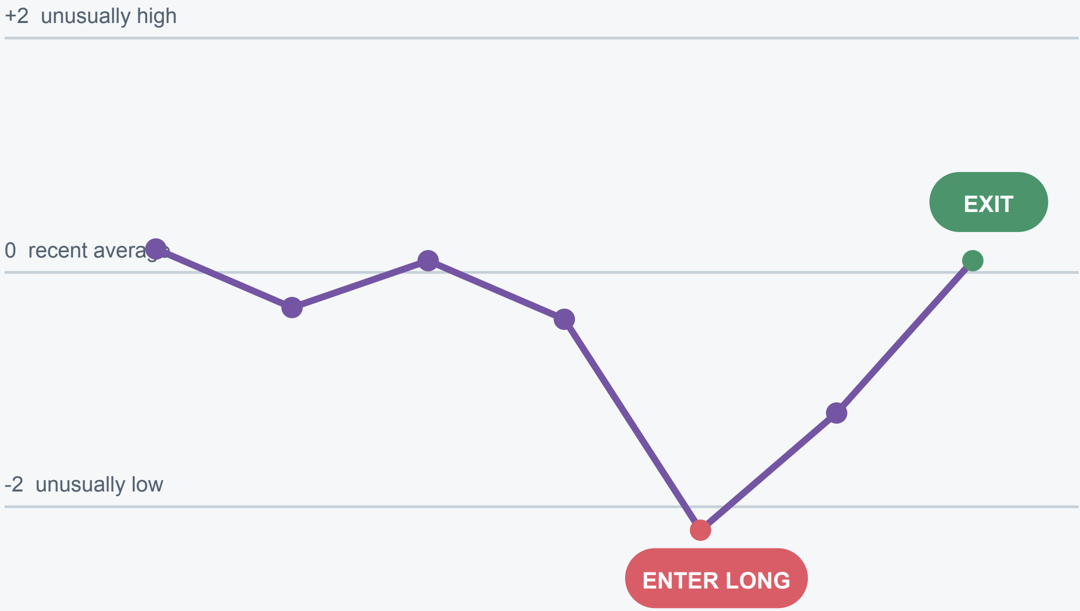
→

EXECUTE
Next open

STRATEGY 2

Mean reversion: spot the stretched spring

Question: is price unusually low compared with its own recent behavior?



Z-SCORE
(price - mean) / std

Memory matters
After entry, stay long until the exit threshold is reached.

Flat prices?
std = 0 -> no signal, not a fake trade.

STRATEGY 3

Momentum: rank, select, hold

Question: which assets have been strongest over the chosen lookback period?

TRAILING RETURN RANKING

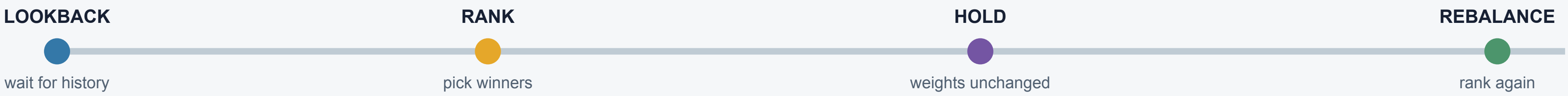
TOP 2

AAPL 50%

MSFT 50%

GOOG 0%

REBALANCE TIMELINE



If require_positive_returns=True and every return is negative, SamQuant holds cash instead of choosing the least-bad loser.

INTERVIEW READY

The safety net around the strategies

Know these protections and you can defend Phase 4 clearly.

NO FUTURE DATA

Past signals stay unchanged when a future price changes.

NEXT-OPEN TRADE

Monday close signal can first trade Tuesday morning.

WARM-UP

Indicators stay inactive until enough history exists.

ALIGNED DATES

Every symbol must describe the same timestamps.

VALID WEIGHTS

Long-only allocations stay between 0% and 100% total.

DETERMINISTIC

Synthetic data and stable tie-breaking repeat exactly.

Your 15-second explanation: Phase 4 contains three configurable strategies that transform validated historical closes into target portfolio weights. The Phase 3 engine shifts those targets and executes them at the next open, keeping research separate from realistic execution.

INTERVIEW CHALLENGE

1. Why weights, not orders?
2. How is look-ahead blocked?
3. Why is mean reversion stateful?
4. Why can momentum hold cash?
5. Why must dates align?
6. What survivorship risk remains?

Verified:
39 automated project tests pass